Pilot Study

Pilot and Feasibility Study on Elderly Support Services Using Communicative Robots and Monitoring Sensors Integrated With Cloud Robotics

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ABSTRACT

Purpose: This pilot before-after study investigated the possible effects of communicative robots, used with a sensing system supported by cloud robotics, in caring for elderly people.

Methods: Two elderly women in nursing homes and 4 care workers participated in the trial. The overnight life rhythm assessments of the study participants and care workers were surveyed to determine when and how the robots should be integrated into care. The system consisted of the robot Sota, a noncontact vital sensor and a sheet-shaped bed sensor. Real-time sensing data and conversations between the participants and robots were sent to the servers, prompting a quick verbal response by the robot supported by cloud robotics.

Findings: Care workers devoted 3 h to the maintenance of records during their most stressful periods. Automatic recording of vital information using robot sensors can improve the quality of nursing care work. Care workers' stress levels were maximized when responding to nurse calls. Temporary responses to nurse calls by the robots may help to effectively reduce the burden on nursing care workers. Robots can stimulate elderly people to communicate more with others ($P < 0.05$). Appropriate vocalization by communicative robots may prevent the deterioration of quality of life in elderly individuals.

Implications: Communicative robots, used with a sensing system, may stimulate elderly people to activate a communication link with others and help care workers to effectively reduce the burden during the night shift. A follow-up study involving a broader research program on communicative robots and elderly care would be beneficial. (*Clin Ther*. 2020;42:364–371) © 2020 Elsevier Inc. All rights reserved.

Key words: activities of daily living, cloud robotics, communicative robot, elderly care, robotics utilization, support services.

INTRODUCTION

Although preliminary research on communicative robots for use by elderly individuals, which has been performed for >10 years, indicates that robots are effective psychosocial aids[1,2] and that their introduction is beneficial for those with dementia,[3] only a few reports have found substantial evidence of this.[4,5] Furthermore, few studies have reported on the potential of communicative robots to reduce the labor burden on nursing care workers.[6]

Japan is not the only country where an aging population is a major social issue. The old-age dependency ratio is estimated to increase in a number of developed countries.[7] Research on the use of robots to care for elderly individuals is attracting attention worldwide.[8,9] Communicative robots were initially thought to be an alternative to pets,[10] but in this age of telemedicine and cloud computing,[11] a new concept in nursing care with communicative robots is being explored. The new technologies are expected to be useful even for dementia population with various stages.[12] The next generation of

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communicative robots should function as part of an entire system's voice-image input/output (I/O) interface that monitors users by using their own intelligence based on artificial intelligence (AI) derived from cloud computing.[13]

The purpose of this study was to identify the tasks and problems associated with the deployment of communicative robots and a monitoring system for medical care in facilities as well as patients' homes because this technology has been discussed as a means of addressing the increased elderly care demand and the anticipated caregiver shortage in Japan. In this study, the robot acts as an intelligent I/O device combined with a remote monitoring system that is connected to a cloud robotics network base supported by learning AI (Figure 1).

PATIENTS AND METHODS

This pilot before-after trials using 2 volunteer elderly people with mild dementia and 4 care workers as study participants was conducted from April to June 2015. The elderly study participants, who participated in the pilot study willingly with written informed consent to participate, included 2 nursing home residents of the Tokyo Seishinkai Flora Tanashi. Patient 1 was a 104-year-old woman (nursing care level 4, bedridden level A2, dementia elderly persons daily life independence level IIa). Patient 2 was an 87-year-old woman (nursing care level 4, bedridden level A2, dementia elderly persons daily life independence level IIa). Four caregivers, who took care of the 2 elderly patients and provided written consent to participate in the study, were also part of the study (3

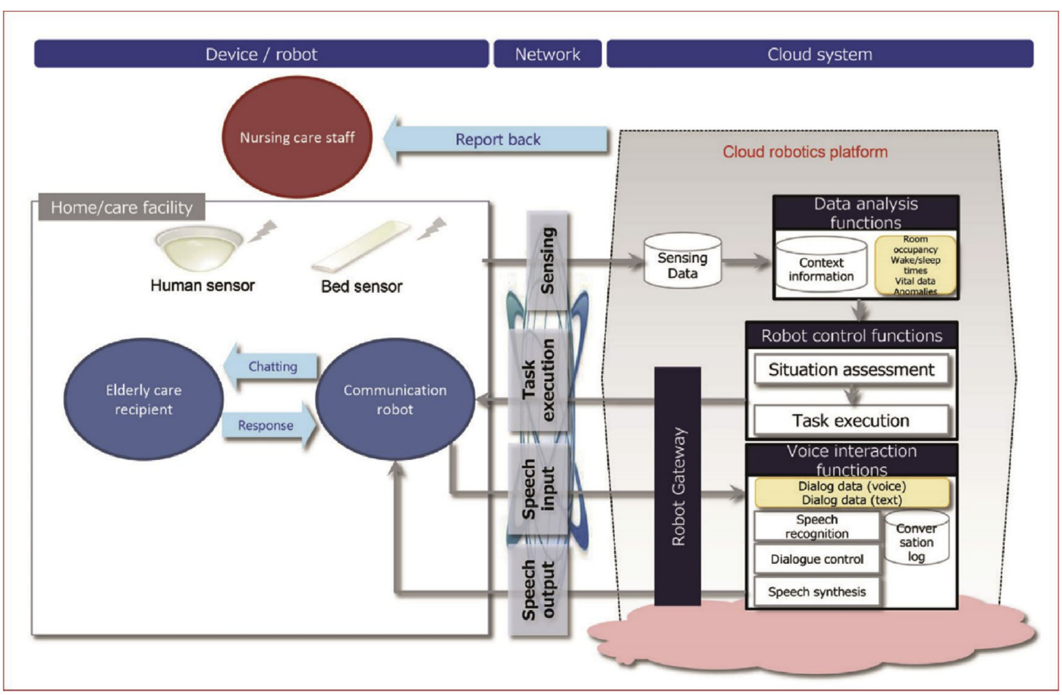

Figure 1. Overview of the sensor system for monitoring elderly individuals using a cloud system[10,13] and an illustration of the elderly monitoring sensor and communicative robot system. Real-time sensing data obtained by vital and bed sensors and the conversations between the participant and the robot are sent to the cloud robotics platform via data input/output devices, where they are subsequently analyzed. While referring to the context information that has been integrated, the state of the participant is analyzed using the sensing data. Dialogue data obtained from the input/output devices of the robot are analyzed by referring to prior speech log analysis, followed by a dialogue control process, resulting in speech synthesis. The final output is sent to the facility staff.

women and 1 man; mean age, 39 years; mean length of caregiving experience, 6.5 years).

The Japanese long-term insurance scheme (designed to address Japan's extreme-aging society in which 3 working-generation people support 2 retired people) classifies applicants according to their need, which is categorized into 7 levels: support levels 1 and 2 and care need scales 1 to 5. Support levels 1 and 2 mean that formal care is not required but some assistance with daily living activities is needed. On the other hand, care need scales 1 to 5 signify a need for formal care. The 5 scales indicate the degree of formal care needed, with 1 indicating the least severe and 5 the most severe.[14] The Degree of Daily Life Independence Score for People With Dementia[15] has 7 scales and is widely used in Japan to evaluate dementia. The assessment criteria and examples of symptom and behavior are given in Supplemental Table I.

The life rhythm assessment was used to observe the living conditions of the residents, and the work content observation record and activity records were used to analyze the current status of late-night work. The Japanese version of the Profile of Mood States (POMS)[16] was used to quantitatively investigate the participants' psychological stress, such as tension, depression, anger, vigor, fatigue, and confusion. A subjective examination[17] by the Industrial Fatigue Study Group of the Japan Society of Industrial Hygiene was used to investigate sleepiness, anxiety, discomfort, fatigue, and fuzziness of the study participants. We also investigated the frequency of nurse calls, which is considered to be a major cause of high workload during midnight shifts.

On the basis of these life rhythm assessment preliminary surveys during April and May 2015, the robot's program to make better verbal communication with the patient was modified by adjusting the interactive function of the communicative robot and improving the sensing technique. After this modification, these robots and sensors were introduced to the nursing environment, and their effect on the elderly patients and the caregivers was evaluated from May to June 2015. Figure 1 shows a schematic of the sensor and communicative robot system.[18] Some of the responses were supported by a cloud robotics base.

The system used in this study consisted of the following: the communicative robot was a Sota (Vstone Co Ltd, Osaka, Japan), the sensor was a noncontact human vital sensor (Mio Corporation Co Ltd, Yokohama, Japan) with a sheet-shaped sleep sensor NemuriSCAN (Paramount Bed Co Ltd, Tokyo, Japan),[19] and the network Wi-Fi in the facility was set up using 3 servers, which enabled information transmission. The Sota (Figure 2) was 280 (height) × 140 (width) × 160 (diameter) mm, weighed 763 g, and had a total of 8 *df* (body: 1 axis; arm: 2 axes × 2; neck: 3 axes). The central processing unit had an Intel Edison module. The I/O system included a camera, monaural microphone, speaker, and light emitting diodes (2 eyes and 1 mouth). Two Wi-Fi, Bluetooth, and USBs were available as interfaces, connecting the robot and AI teacher.

The effect of communicative robots and sensors on the patients and caregivers was compared by t-teat using EZR (Saitama Medical Centre, Jichi Medical University, Saitama, Japan). $P < 0.05$ was considered statistically significant.

The study was reviewed and approved by the ethics committee of the Tokyo Seishin-kai, Tokyo, Japan, a social welfare corporation (TS2014-001).

RESULTS

Lifestyle Patterns and Behavioral Motivation Analysis of Elderly Nursing Home Residents

The lifestyle patterns of the 2 elderly patients were surveyed during a 4-day period, mainly during the night (16:30 to 09:30, midnight working hours) using a life rhythm monitoring sheet at 15-min

Figure 2. Communicative robot (Sota).

intervals. The participants spent >4 h in a common living room and dining room. They spent only 18%–19% of their time in their room (ie, just >1 h; Supplemental Figure 1).

According to an analysis of behavioral motivation (Table I) during the presurvey period, the frequency of conversation between patient 1 and care staff seemed to change in accordance with the frequency of movement. The amount of conversation between the residents increased on days when the frequency of movement was high (Supplemental Figure 2). There were differences in the daily activities of the 2 patients. Each patient had a different life pattern, indicating the need for a tailor-made care program.

Analysis of Psychological Burden and Fatigue of Caregivers

The excessive burden of nursing care has been a serious problem during night shifts. The survey of the rate of psychological burden and fatigue on 4 care workers based on the POMS and the subjective examination findings that were recorded every hour during the night shift (Figure 3) indicated that fatigue peaked just before a nap (01:00 to 03:00) and increased again from dawn to early morning.

Of the total working time, the recording was taken for >3 h (mean [SD], 181 [24] minutes) of a total of 15 h. The recording period was concentrated just before the nap and during peak fatigue in the early morning. The time required for excretion assistance, which was generally requested by nurse call, was also >3 h. The nurse call frequency seemed to peak before naps and early in the morning (Figure 3C). The frequency correlated with some subjective feelings (vigor, depression, anger, fatigue, and confusion) of care workers during the night shift (Supplemental Figure 3). The mean POMS score tended to increase with the increasing frequency of the nurse calls (quadratic polynomial regression $r = 0.49$; Supplemental Figure 4). These results suggest that the increased nurse calls enhanced the psychological burden.

Sensors and Robots

Technical adjustments on the communicative robot were performed in parallel with a field study of the nursing care service. Adjustments and improvements were made to the dialogue function and sensing function. With regard to I/O, the sound collection device and speaker had to be positioned in suitable locations because the participants had reduced hearing and speech abilities. With improved hearing and sensing abilities (ie, recognition accuracy), the communicative robot could interact with the participants in real time.

It was possible to automatically record the time the participants stayed in bed, the time they fell asleep, the time they woke up, the number of times they woke up during the night, sleeping time, respiratory frequency, and the reason for waking up by using sensors, such as a noncontact vital sensor and NemuriSCAN (Supplemental Figure 5). These sensors reduced the burden on caregivers by providing real-time vital information about the patients at night. On the basis of a predeveloped scenario, the communicative robot used the information obtained from the sensor as a trigger point to make appropriate calls to elderly

Table I. Number of behavioral motivation analyses of participants during a 4-day period (including conversations with staff, conversations between residents, smiles, and amount of movement).

Patient	Day 1	Day 2	Day 3	Day 4	Mean (SD)
Patient 1					
Conversation with staff	6	7	7	25	11.3 (9.2)
Conversation with resident	1	0	2	3	1.5 (1.3)
Smile	0	0	0	1	0.3 (0.5)
Patient 2					
Conversation with staff	16	11	16	15	14.5 (2.4)
Conversation with resident	10	9	10	12	10.3 (1.3)
Smile	9	4	7	14	8.5 (4.2)
Movement	15	15	12	18	15.0 (2.4)

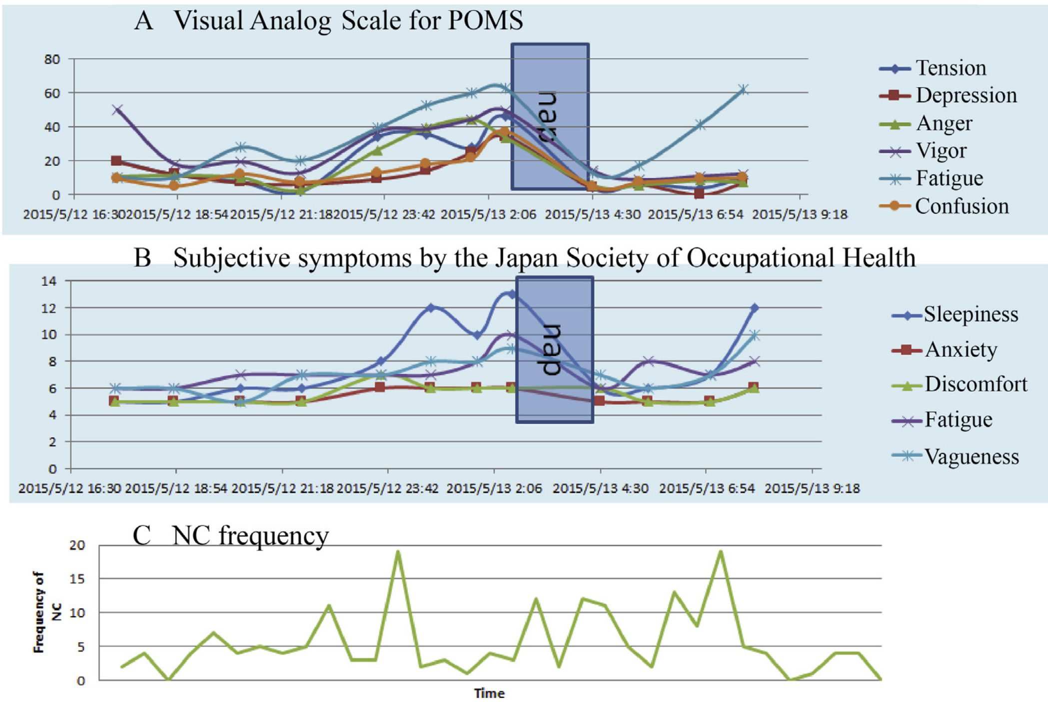

Figure 3. Results of Profile of Mood States (POMS) (A), subjective examination (B), and nurse call (NC) frequency (C) per hour during night shifts.

patients according to a time schedule based on the patients' daily routine (Figure 4).

Conversations with the robot, which were activated by the sensor, were useful for safety confirmation. There was a case in which an error in taking pills was avoided by a conversation on ingestion confirmation. Information on nocturnal life, which is usually difficult to observe, was also obtained using this system. In some cases, the robot's voice guided the movement of the participants and prevented unexpected falls. Table II gives the number of conversations with patient 1 after the introduction of the system. Compared with the case before the introduction of the system (Table I), the number of conversations with care workers increased from a mean of 11.3–14.0, and conversations with residents increased significantly from 1.5 to 9.5 ($P < 0.05$). Conversations with the robot increased from 4 times on the first day to 8 times on the fourth day. The robot, triggered by sensor information, spoke to the participant a mean of 5 times a day.

DISCUSSION

Given that this is a pilot and feasibility study with a small number of participants and limited equipment, it is important to mention that concrete conclusions cannot be obtained based on this study alone; however, there are many insights from this study that can be applied by researchers and medical care facilities. The study results suggest the following. First, communicative robots can stimulate elderly individuals to activate a communication link with others and participate in society. Second, the temporary response to nurse call by the robots may help to effectively reduce the burden on nursing care workers during the night shift. Third, automatic recording of vital information using robot sensors can improve the quality of nursing care work. Fourth, appropriate vocalization by communicative robots with sensors may prevent the deterioration of activities of daily living and initiate dialogue. Fifth, the positive effect of communicative robots detected among older people with the moderate level of

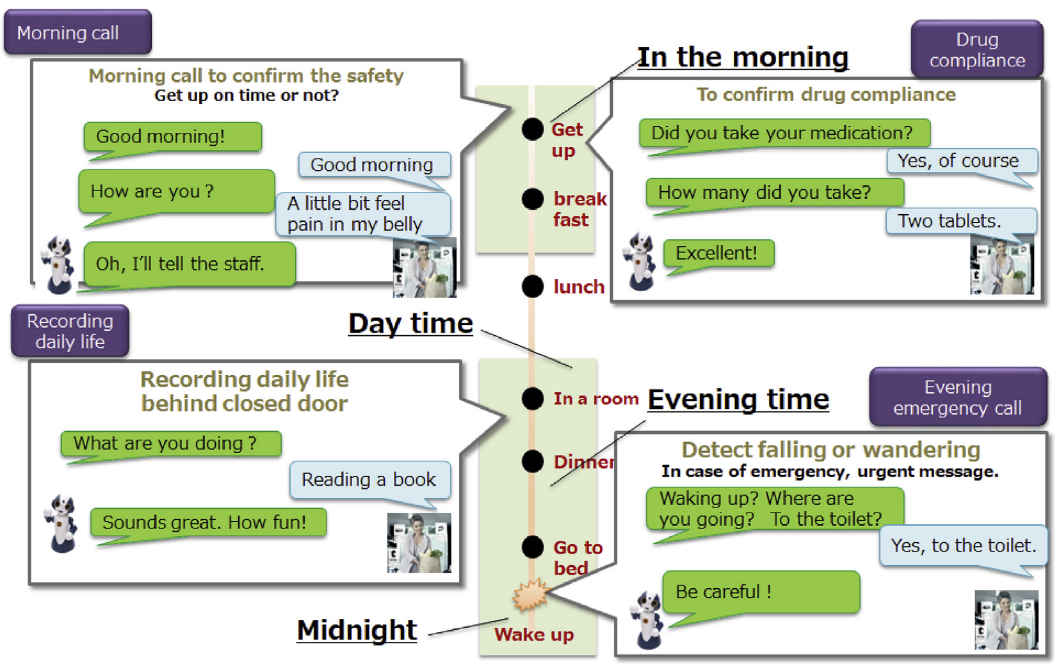

Figure 4. Conversation between the robot and participants. The conversations are triggered by sensors. A morning call was used to confirm the safety of the elderly patient. Conversations to confirm drug adherence prevented elderly patients from taking the wrong medicine or doses. Conversations in the living room made it possible to record the daily life of the elderly patient behind closed doors. Midnight watching and calling by the communicative robot were used to detect risks of falling or wandering.

Table II. Number of conversations of patient 1 after the introduction of the robot.

Patient 1	Day 1	Day 2	Day 3	Day 4	Mean (SD)
Conversation with staff	14	18	7	17	14.0 (5.0)
Conversation with residents	6	13	5	14	9.5 (4.7)
Conversation with robot	4	7	7	8	6.5 (1.7)

dementia will be welcomed by care professionals working in nursing facilities and in the communities, where new technologies are expected to be used for the population with various stages of dementia.[20] Sixth, also in this study, care professionals were initially not so certain about the robot's capabilities. However, they gradually began to accept the robots as their assistants, making the best of what they could.

Nevertheless, many problems have yet to be resolved for this system. Regarding the management of nursing work, it is important to protect the privacy of elderly individuals, and data security is fundamental. Security is multidimensionally built by physical, platform, network, proactive risk, and compliance with an authentication system. In addition, security should be always revised. If care workers do not follow these guidelines, they cannot participate in this field. It is also essential to create an appropriate scenario that matches the present state of nursing welfare facilities and custom-made programs for every elderly person, and it is necessary to establish an effective method to introduce robots into the care environment by analyzing the characteristics of nursing care work. These tasks remain important when introducing this system into a home in the future.

In addition, in terms of technical aspects, such as robot tuning, it is essential to ensure the secure handling of sensing data, given the possibility of its use in home health care. The sensor system should protect the dignity of the participants. Further improvement of the speech engine, especially language structural analysis and speech analysis, is required.

The role of communicative robots for care workers is considered to improve care work at the sociopsychological level,[21] including reducing stress,[22] relieving loneliness, and improving communication among people who require care. However, this method alone will not solve the challenges with the care situation that will arise in Japan by 2040, an incredible super-aging society with 3 working-generation people (from 15 to 65 years old) having to support 2 elderly people (>65 years old).[23]

The nursing care field needs to improve the performance of the elderly individuals living at home or in facilities via conversations with robots and by encouraging them to participate. Even more necessary, however, is a cloud-based safety integrated robot system[24] equipped with sensors to detect risks in daily life (eg, unintentional falls)[25] before they occur.

The system will support home and remote medical care through learnable AI. To achieve this, a revolutionary leap in hardware and software is required, such as improvements in the I/O capability of the robot, acquisition of object tracking and movement capabilities,[26] improvement in the intelligence level, design of cloud robotics to integrate sensors and other robots, and the development of a management response system supported by AI that can learn crisis management algorithms.

CONCLUSIONS

Communicative robot and monitoring sensor systems are within the reach of elderly patients and care workers. Whether at a care facility or at home, information and communications technology is essential for providing care to elderly individuals in the future. Local governments, companies, researchers, engineers, caregivers, and elderly participants need to collaborate to develop a communicative robot. A new Japan-wide field research program on communicative robots and elderly care using >1000 robots is being planned in cooperation with all stakeholders in Japan. We will participate in this program by using 110 robots based on the results of this pilot study.

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DISCLOSURES

The authors have indicated that they have no conflicts of interest regarding the content of this article.

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REFERENCES

1. Shibata T. Therapeutic seal robot as biofeedback medical device: qualitative and quantitative evaluations of robot therapy in dementia care. *Proc IEEE*. 2012;100:2527–2538.
2. Robinson H, Macdonald B, Kerse N, Broadbent E. The psychosocial effects of a companion robot: a randomized controlled trial. *J Am Med Dir Assoc*. 2013;14(9):661–667.
3. Tapus A, Tapus C, Mataric MJ. The use of socially assistive robots in the design of intelligent cognitive therapies for people with dementia. In: *Proceedings of the IEEE International Conference on Rehabilitation Robotics, ICORR, Japan*. 2009: 924–929.
4. Bemelmans R, Gelderblom G, Jonker P, De Witte L. Socially assistive robots in elderly care: a systematic review into effects and effectiveness. *J Am Med Dir Assoc*. 2012;13(2):114–120.

5. Rantanen P, Parkkari T, Leikola S, Airaksinen M, Lyles A. An in-home advanced robotic system to manage elderly home-care patients' medications: a pilot safety and usability study. *Clini Ther*. 2017;39(5):1054–1061. https://doi.org/10.1016/j.clinthera.2017.03.020.
6. Masuyama S, Obayashi K, Ogata T, Kondo H, Okamoto Y, Ishii Y. Measuring impacts of introducing communicative robots with infrared radiation monitoring system on workload of night shifts in nursing care facilities. *Age and Ageing*. 2017;46(3). iii13-iii59.
7. Demographic Profiles: Citation: United Nations, Department of Economic and Social Affairs, Population Division. In: *World Population Prospects: The 2017 Revision, Volume II: Demographic Profiles. ST/ESA/SER.A/400*; 2017. https://population.un.org/wpp/Publications/Files/WPP2017_Volume-II-Demographic-Profiles.pdf. Accessed April 4, 2019.
8. Obayashi K, Kodate N, Masuyama S. Socially assistive robots and their potential in enhancing older people's activity and social participation. *J Am Med Dir Assoc*. 2018;19(5):462–463.
9. Robo M.D.; a home care robot for monitoring and detection of critical situations. In: *Conference: ECCE 2010 - European Conference on Cognitive Ergonomics, Delft, The Netherlands, August 25-27, 2010, Proceedings of the 28th Annual Conference of the European Association of Cognitive Ergonomics*. January 2010. https://doi.org/10.1145/1962300.1962391.
10. Wada K, Shibata T, Musha T, Kimura S. Robot therapy for elders affected by dementia. *IEEE Eng Med Biol Mag*. 2008;27:53–60.
11. Goldberg K, Kehoe B. *Cloud Robotics and Automation: A Survey of Related Work. Technical Report UCB/EECS-2013-5*. Berkeley: EECS Department, University of California; 2013.
12. Jones C, Sung B, Moyl W. Assessing engagement in people with dementia: a new approach to assessment using video analysis. *Arch Psychiatr Nurs*. 2015;29:380.
13. Kagaya T. Strategy and efforts for robotics integration aiming at combining information and communications technology with robots. *NTT Tech Rev*. 2012;10(11):1–7.
14. Moriyama Y, Tamiya N, Kawamura A, Mayers TD, Noguchi H, Takahashi H. Effect of short-stay service use on stay-at-home duration for elderly with certified care needs: analysis of long-term care insurance claims data in Japan. *PLoS ONE*. 2018;13(8), e0203112. https://doi.org/10.1371/journal.pone.0203112.
15. Ministry of Health, Labour and Welfare. *Degree of Daily Life Independence Score for People with Dementia [Ninchishō Kōreisha No Nichijōseikatsu Jiritsudo]*; 2013 (Japanese) https://www.mhlw.go.jp/topics/2013/02/dl/tp0215-11-11d.pdf. Accessed March 17, 2019.
16. Arruda JE, Stern RA, Somerville JA. Measurement of mood states in stroke patients: validation of the visual analog mood scale. *Phys Med Rehabil*. 1999;80(6):676–680.
17. Kubo T, Tachi N, Takeyama H, et al. Characteristic patterns of fatigue feelings on four simulated consecutive night shifts by "Jikaku-shoshirabe". *Sangyo Eiseigaku Zasshi (J Occup Health)*. 2008;50:133–144.
18. Watanabe S. Implementation and application of robot information processing functions—using communicative robots for elderly care support. *New Breeze Autumn*. 2015;10–13.
19. Kogure T, Shirakawa S, Shimokawa M, Hosokawa Y. Automatic sleep/wake scoring from body motion in bed: validation of a newly developed sensor placed under a mattress. *J Physiol Anthropol*. 2011;30(3):103–109.
20. Obayashi K, Kodate N, Masuyama S. Do age, gender and stage of dementia matter in the intervention study of communicative robots? *AGE&Aging September*. 2018. https://doi.org/10.1093/ageing/afy140.18.
21. Broekens J, Heerink M, Rosendal H. Assistive social robots in elderly care: a review. *Geron Technol*. 2009;8(2):94–103.
22. Kanoh M, Oida Y, Nomura Y, et al. Examination of practicability of communicative robot-assisted activity program for elderly people. *J Robotics Mechatron*. 2011;23(1):3.
23. Population Projections for Japan: 2016-2065 National Institute of Population and Social Security Research. Tokyo, Japan. Available online: http://www.ipss.go.jp/pp-zenkoku/j/zenkoku2017/pp29_ReportALL.pdf.
24. Manzi A, Fiorini L, Esposito R, et al. Design of a cloud robotic system to support senior citizens: the KuBo experience. *Autonomous Robots*. 2017;41(3):699–709.
25. Sumiya T, Matsubara Y, Nakano M, Sugaya M. A mobile robot for fall detection for elderly-care. *Proced Comput Sci*. 2015;60:870–880.
26. Dorronzoro Zubiete E, Nakahata K, Imamoglu N, et al. Evaluation of a home biomonitoring autonomous Mobile Robot. *Comput Intell Neurosci*. 2016;2016:1–8.

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APPENDIX A. SUPPLEMENTARY MATERIALS

Table S1. Degree of Daily Life Independence Score for People with Dementia (DDLIS-PD).

Scale		Assessment criteria	Examples of symptom and behaviour
I		Signs of mild form of dementia, but ADL is possible in their homes and community	
II		Many symptoms/behaviours or difficulties in communication that may interfere with ADL, independence can be achieved with some assistance	
	a	Above-mentioned symptoms/behaviours observed outside their homes	Difficulties with navigating in the street, daily activities such as shopping, office work, financial management.
	b	Above-mentioned symptoms/behaviours observed in their homes	Difficulties with medication management, answering the phone, waiting alone in the house.
III		ADL is declining, and occasional nursing care is necessary. Symptoms, behaviour, and difficulties are seen in communication that cause problems in daily life.	Difficulties with changing clothes, eating, toileting, mismanagement of fire, sexual abnormality, and unusual behaviours (e.g., putting things into the mouth, pick up things aimlessly, shouting)
	a	Above-mentioned symptoms/behaviours observed primarily during daytime	
	b	Above-mentioned symptoms/behaviours observed primarily during night time	
IV		Constant care is needed. Symptoms, behaviours and difficulties in communication often cause problems for ADL.	Similar to III
M		Almost impossible to communicate when trying to meet the needs of ADL. Marked mental symptoms and behavioural problems or serious physical problems are seen, and specialised care is necessary.	Mental symptoms such as delirium, delusions, excitement, self-harm, etc. Continued problematic behaviours caused by mental symptoms.

(Translated by the authors).

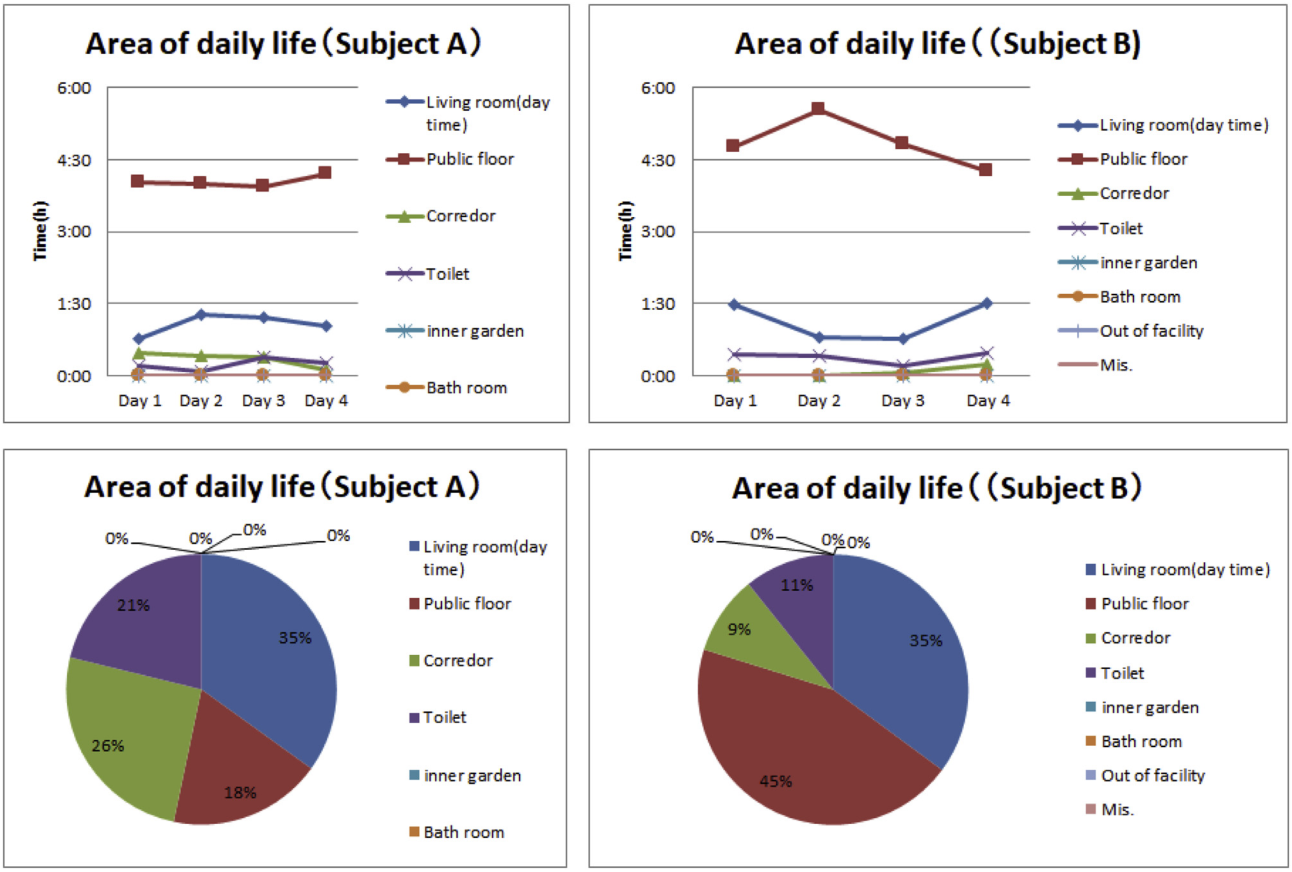


Figure S1. The lifestyle patterns of Subject A and B.

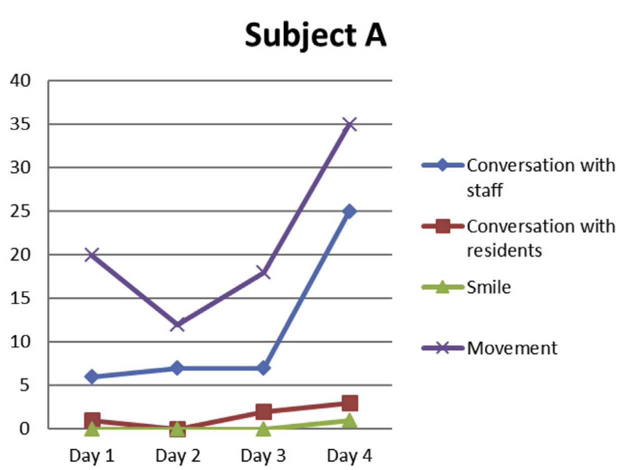


Figure S2. Behavioural motivation analysis of the number of conversations vs. frequency of movement.

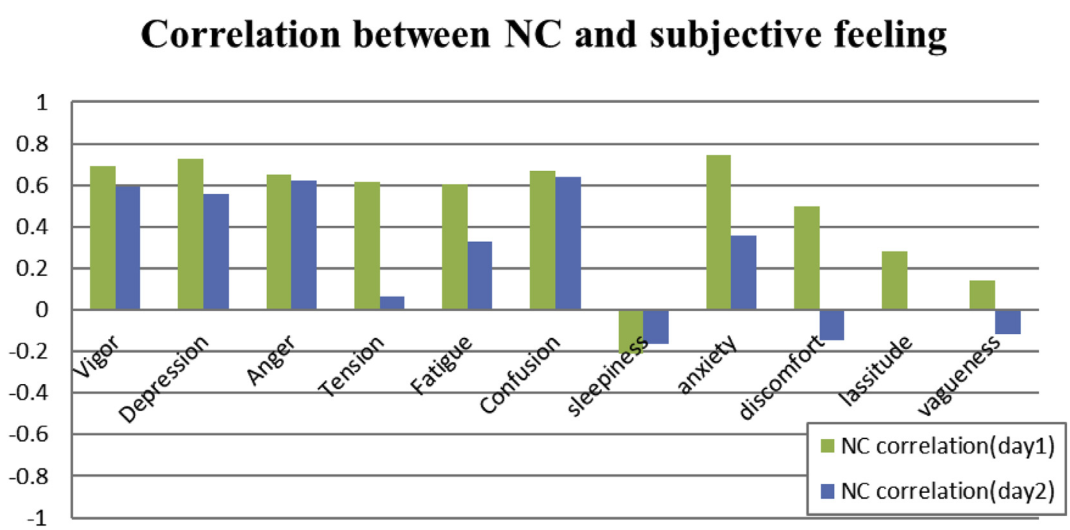


Figure S3. Correlation between nurse call frequency (NC) and subjective feelings of a care worker during night shift.

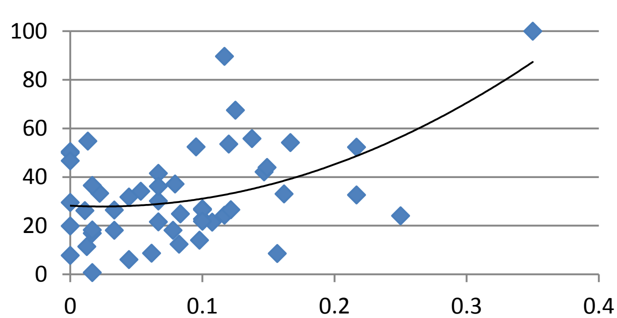
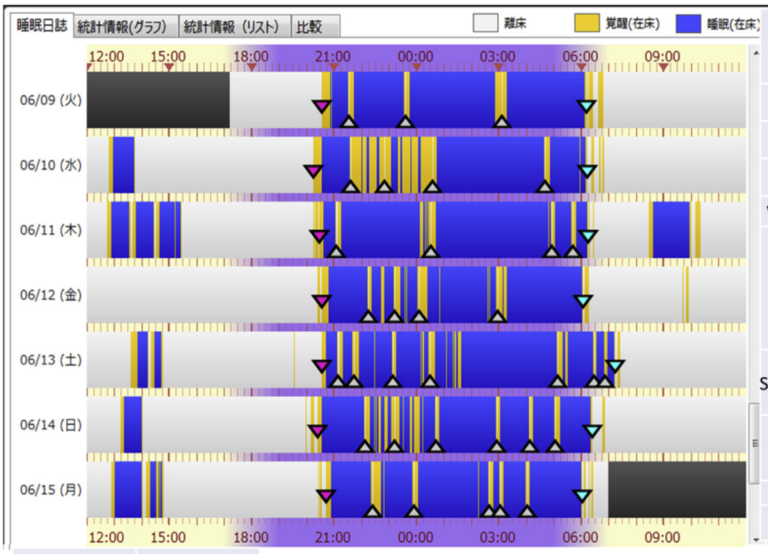


Figure S4. Relationship between nurse call frequency (X-axis) and POMS (Y-axis, average).



Date		1st	2nd	3rd	4th
		June 9	June 11	June 15	June 19
Go to bed		20:35	20:29	20:44	20:14
Falling asleep		20:11	20:38	20:55	20:33
Getting up		6:12	6:16	5:58	5:54
Wake up during sleep		3	4	5	4
Wake up time	1st	21:32	21:04	22:24	22:00
	2nd	23:36	0:29	23:54	1:45
	3rd	3:06	4:53	2:36	4:44
	4th		5:41	3:00	4:55
	5th			4:00	5:18
Sleep duration（min）		501	512	478	438
In bed time（min）		570	577	552	553

▽ (purple)	Falling asleep
▽ (awua)	Getting up
△	Waking up
blue	sleep (in bed)
yellow	awake (in bed)

Figure S5. A sample views of sleep/awake status by NemuriScan.